Design Reflection: Date Display GUI

1. Main Functionalities

This application is a Date Display GUI that allows users to change dates easily. When the program starts, it automatically shows the current date in the format dd/MM/yyyy. Users can enter an integer value x into the input text field and use the buttons to move the displayed date by x days or x months. The application supports moving to x days before, x days after, x months before, and x months after the currently displayed date. It provides a simple and clear interface for date adjustment.

2. Design Decisions & Rationale

The main design decision for this assignment is the use of the Model-View-Controller (MVC) pattern. The system is divided into three main classes to follow the Separation of Concerns principle. The DateModel class manages the date data and calculations. The DateView class handles the user interface using Java Swing. The DateController class controls the application by adding listeners to the View, updating the Model based on user input, and updating the View display. This design makes the code easier to read, maintain, test, and modify.

3. Brief Explanation of the Implementation

In the Model, Java's java.time.LocalDate class is used to manage date calculations safely. This helps handle cases such as leap years, February 29th, and different month lengths without complex conditional checks. The Controller uses a private inner class named ActionHandler, which implements ActionListener, to process button clicks. It also checks user input for empty strings, negative numbers, very large numbers, and invalid characters. If an error occurs, the Controller catches the exception and asks the View to display an error dialog using JOptionPane. This prevents the application from crashing.